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# Electronic health records and polygenic risk scores for predicting disease risk

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Abstract | Accurate prediction of disease risk based on the genetic make-up of an individual is essential for effective prevention and personalized treatment. Nevertheless, to date, individual genetic variants from genome-wide association studies have achieved only moderate prediction of disease risk. The aggregation of genetic variants under a polygenic model shows promising improvements in prediction accuracies. Increasingly, electronic health records (EHRs) are being linked to patient genetic data in biobanks, which provides new opportunities for developing and applying polygenic risk scores in the clinic, to systematically examine and evaluate patient susceptibilities to disease. However, the heterogeneous nature of EHR data brings forth many practical challenges along every step of designing and implementing risk prediction strategies. In this Review, we present the unique considerations for using genotype and phenotype data from biobank-linked EHRs for polygenic risk prediction.

Healthcare has long pursued an understanding of the personal risk factors that contribute to disease onset. Accurate estimation of disease risk can not only improve our understanding of the associated factors underlying disease development but also provide early warning for patients at increased risk of developing certain diseases. This knowledge could help guide clinical decisions, such as prescribing pre-emptive lifestyle changes or therapies for those at greater risk of disease.

For a long time, epidemiologic studies have served as a means to establish links between individuals' demographics, lifestyles and environmental exposures and their disease risks. For example, the pooled cohorts equation estimates the 10-year risk of coronary heart disease using an individual's age, sex, race, total and high-density lipoprotein cholesterol levels, smoking status, diabetes mellitus status, systolic blood pressure level and hypertension therapy status[1]. Genetic risk profiling — the identification of high-risk individuals based on analysis of genetic variations statistically associated with disease — was initially focused predominantly on estimating disease risks for familial diseases, notably breast and ovarian cancer, and Mendelian disorders, such as cystic fibrosis. Mutations in *BRCA1* and *BRCA2* are strong genetic factors that determine the risk of developing breast and ovarian cancer in women by the age of 70 years[2]. Similarly, polymorphisms in the *CFTR* gene sequence are highly predictive of cystic fibrosis[3].

Over the past two decades, the investigation of genetic variation underlying disease susceptibility has increased considerably. Most notably, genome-wide association studies (GWAS) have investigated tens of millions of single-nucleotide polymorphisms (SNPs) for associations with complex diseases. However, results from numerous GWAS have revealed that the majority of statistically significantly associated genetic variants have small effects[4] and may not be predictive of disease risks[5], and many diseases are associated with tens of thousands of genetic variants[6]. These findings have led to the resurgence of the polygenic risk score (PRS), an aggregate measure of many genetic variants — that is, risk factors — weighted by their individual effects on a given phenotype.

Although disease risk prediction using a PRS is at an early stage, it is starting to be applied to many common diseases. For example, the Psychiatric Genomics Consortium has led efforts for many early PRS-related research studies[7,8], and PRSs have been significantly associated with the risk of schizophrenia[8], antipsychotic drug treatment[9] and psychotic experiences in adolescents[10]. These studies and others demonstrate the potential of assessing disease risk using a PRS in population-based studies. However, epidemiologic studies are expensive and complex to run, which raises the question of whether a PRS could be developed and applied in a clinical setting using genetic data that are more readily available.

Electronic health records (EHRs) have been adopted worldwide to improve health outcomes by enhancing the delivery, efficiency, quality and safety of healthcare while safeguarding the privacy of patients. The aim of EHRs is to reduce health disparities and the cost of healthcare, engage patients in the healthcare process and improve public health. A growing number of EHR systems are linking clinical records of patients with their

Genome-wide association studies
(GWAS). Studies in which associations between genetic variation and a phenotype or trait of interest are identified by genotyping cases (for example, diseased individuals) and controls (for example, healthy individuals) for a set of genetic variants that capture variation across the entire genome.

Polygenic risk score
(PRS). A weighted score calculated from numerous genetic variants for predicting disease risk. A PRS is calculated as the sum of risk alleles multiplied by their association coefficients.

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https://doi.org/10.1038/s41576-020-0224-1

Electronic health records
(EHRs). Also known as electronic medical records. Digitally stored patients' medical history.

Biobanks
Repositories that store biological samples, including blood or tissue samples, for research use. Increasingly, the term biobank is used to denote a population cohort study with stored biological samples.

Endophenotypes
The physiological traits that are related to a disease trait; for example, for hypertension this could include blood pressure, angiotensin levels or salt sensitivity.

genetic data, obtained by DNA genotyping or sequencing biospecimens, such as blood or tissue samples, and deposited in biobanks[11–18]. Some of these efforts include, among others, BioBank Japan[17], the National Biobank of Korea[18], China Kadoorie Biobank[19], FinnGen[20], the Precision Medicine Initiative[21] and the UK Biobank project[22]. These rich sets of data, which match patient phenotypes with genotypes, provide a prime opportunity to evaluate genetic risk for various phenotypes simultaneously. However, whether EHR data can be used to answer questions of risk prediction has not been fully answered, despite the fact that these data are currently being used to conduct population-based studies[23].

In this Review, we present new opportunities for developing and implementing PRS predictions using biobank-linked EHR data. Given the heterogeneous nature and complexity of EHR data, we provide our perspectives on some of the unique challenges in developing PRS predictions using EHR data. We conclude by outlining the needed steps to improve the confidence and utility of PRS predictions for susceptibility to common diseases through the use of EHR-linked biobanks. For details on the development[24] and utility[25] of a PRS in disease risk prediction we refer readers to other recent reviews.

## Constructing a PRS from EHRs

For a specific trait or disease, the construction of a PRS is separated into two stages. In the discovery stage, genetic associations are obtained from large-scale GWAS. Statistically, the larger GWAS sample size enables more precise estimation of genetic associations. In the construction stage, a PRS is generated in an independent and generally smaller-sized target population using the GWAS summary statistics, which include the measured effect size and the associated $P$ value for each SNP. For a given phenotype, a PRS is constructed as the sum of multiple SNPs' alleles weighted by their effect sizes. Yet many decisions made during the calculation, such as the SNP ranking method[26,27], selection threshold[28] and linkage disequilibrium adjustment[29,30], can affect the final PRS. Thus, there can be multiple versions of PRSs for one disease (for practical aspects of constructing PRSs, see REF.[31]).

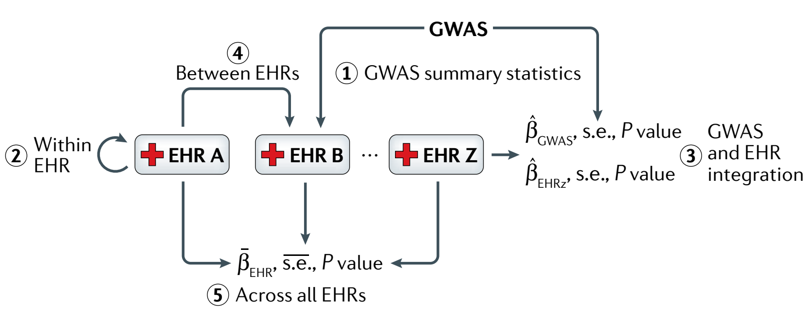


Fig. 1 | **Data integration for a PRS in EHRs.** Electronic health records (EHRs) enable data integration with external sources of information to develop and utilize a polygenic risk score (PRS). External genome-wide association study (GWAS) summary statistics can be applied to target EHR data to generate PRSs (**1**). An EHR can generate its own PRSs and apply it to existing and new patients (**2**). Summary statistics from EHRs can be integrated with external GWAS summary statistics to produce integrated PRSs (**3**). PRSs generated from one EHR can be applied to another EHR (**4**). Multiple EHRs can produce combined PRSs, using distributed algorithms (**5**).

If patients of one biobank-linked EHR system are the target population, a PRS for those patients can be constructed for a particular trait using external GWAS summary statistics for the same trait (FIG. 1). However, only a small percentage of phenotypes in EHRs have corresponding external GWAS summary statistics. Moreover, some EHR systems, such as the UK Biobank, which holds genetic and phenotypic data for approximately 500,000 individuals, rival the cohort size of some GWAS. For such EHR systems, it is feasible and can be advantageous to calculate PRSs using their own patients' data. That is, genotype data for a subset of patients (the training set) can be used to obtain summary statistics of genetic associations for a given phenotype, which can then be used to construct a PRS for the rest of the patients in the same EHR system (the testing set).

Efforts are already underway to use biobank-linked EHR data to construct PRSs. For example, two studies have evaluated PRSs for multiple cancers using the Michigan Genomics Initiative EHR[12,32]. The UK Biobank has also been used to carry out various PRS studies in psychiatric disorders[33] and cancer[34]. Recently, a PRS was used to identify a proportion of the study population with risk equal to monogenic mutations for coronary artery disease, atrial fibrillation, type 2 diabetes mellitus (T2DM), inflammatory bowel disease, breast cancer[35] and obesity[36]. These studies demonstrate that the predictability of a PRS can reach the threshold of clinical significance.

## EHR phenotypes for a PRS

EHRs provide a much wider array of phenotypes compared with traditional population-based studies (FIG. 2). EHRs include all disease diagnoses for each patient who has been seen in a healthcare system as well as their laboratory tests, imaging and diagnostic measures. This rich set of clinical data can be used to better define disease or as endophenotypes[37,38]. In addition to using phenotype information to identify cases of disease and/or endophenotypes, EHRs can be used to develop refined control definitions. That is, healthy controls (patients who have no evidence of disease in their EHRs) can be readily identified. In contrast to traditional population-based studies, which define the variables to be collected before study enrolment and generally rely on individual self-reporting to identify controls, EHR systems enable researchers to select the variables they need from the available data.

When creating disease phenotype definitions to extract data from EHRs, there are several strategies that can be employed[39–41]. Many of these strategies have been used with clinical data from EHRs for many years[42–44]. However, the ability to assign appropriate disease phenotype definitions varies considerably from one trait to another.

***ICD codes.*** International Classification of Diseases (ICD) codes have been the primary source of clinical data used to extract phenotypes from EHRs. In the simplest cases, the number of occurrences of a particular ICD code is used to define the case and control status. However, multiple ICD codes can be used to record the same diagnosis. Hence, different healthcare systems may have different coding practices, which can make

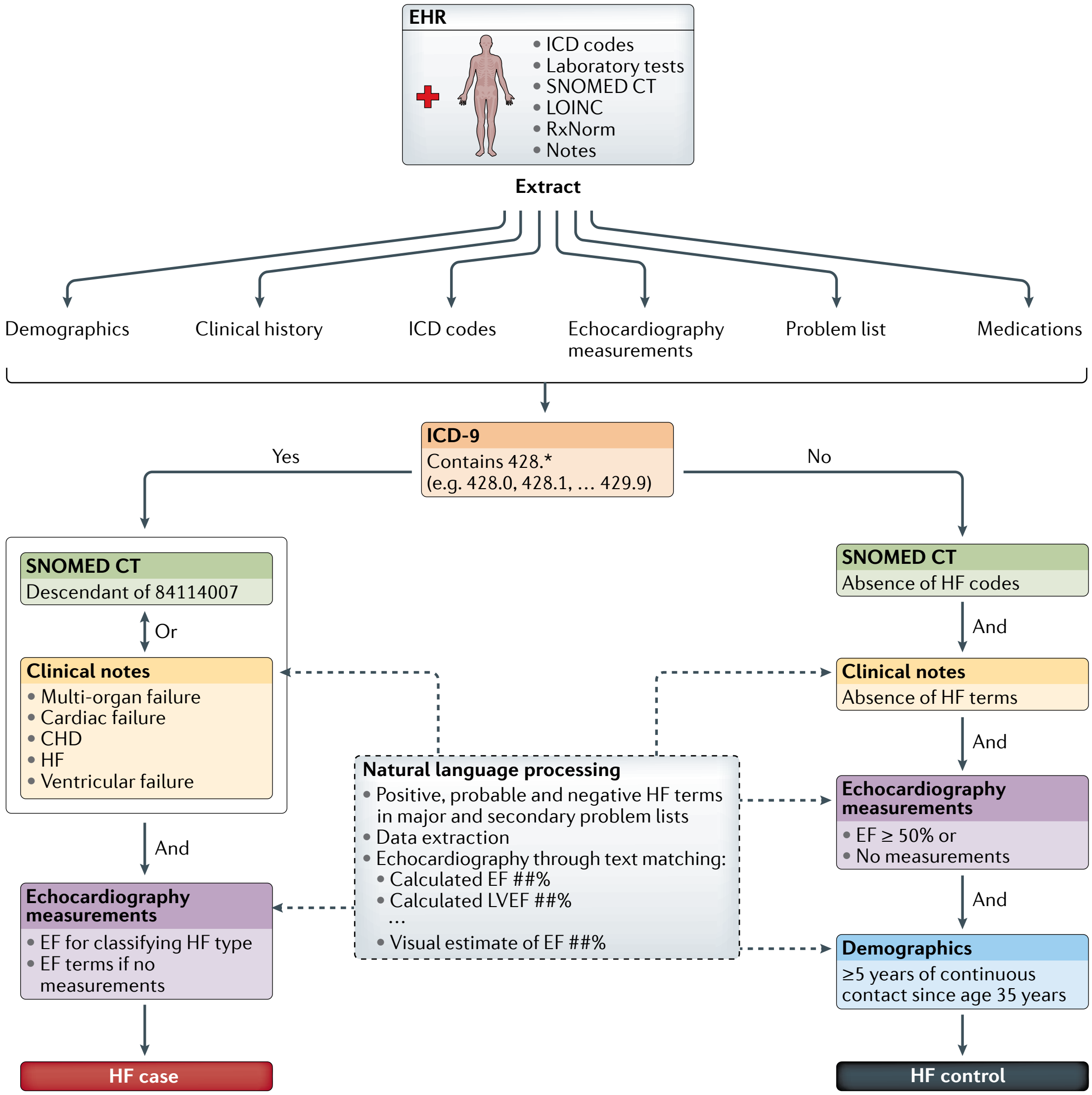


Fig. 2 | **Extracting phenotypes from EHR data for deriving PRSs.** Here, we use an example phenotyping algorithm for heart failure (HF) developed by the Mayo Clinic in the electronic Medical Records and Genomics (eMERGE) project to demonstrate a typical pipeline for electronic health record (EHR) phenotype extraction[111]. An EHR system contains various types of clinical data for each patient, including the International Classification of Diseases (ICD) codes for disease diagnosis[112], laboratory measurements (for example, cholesterol level), Systematized Nomenclature of Medicine –Clinical Terms (SNOMED CT)[113], Logical Observation Identifiers Names and Codes (LOINC) for identifying test observations[114], RxNorm for medications[115] and clinician notes that capture relevant information. In the HF phenotype algorithm, patients' demographics, clinical history, ICD codes, echocardiography measurements, structured and unstructured problem lists, and medications are extracted from the EHR and fed through a rule-based tree to classify patients' disease status. The typical first step is to select patients based on the relevant ICD (ICD-9 or ICD-10) codes (BOX 1). If patients have HF-related ICD-9 codes (428.*), then they will be further evaluated using key terms in the clinical notes, related SNOMED CT codes and echocardiography measurements to determine whether they are cases. If patients do not have HF ICD-9 codes, they can potentially be classified as controls, provided they also pass other criteria specified by the algorithm. A marked portion of the EHR phenotyping algorithm uses natural language processing because of the effectiveness of extracting information from unstructured data. Finally, the HF phenotype obtained from the algorithm can be used for deriving polygenic risk scores (PRSs). ##% represents a regular expression pattern. CHD, coronary heart disease; EF, ejection fraction; LVEF, left ventricular ejection fraction.

combining or comparing EHR data across healthcare systems challenging (BOX 1).

Although ICD codes are convenient to use, the codes themselves are suboptimal to define diseases[45], as the presence of ICD codes primarily indicates billing has occurred rather than true disease diagnosis. Integrating diagnostic codes with clinical laboratory measurements, medications and clinical notes in electronic phenotype algorithms can increase phenotype sensitivity and specificity[46].

***Clinical notes.*** Detailed information about patients is often present in the notes taken by clinicians. These notes can be processed manually or using natural language processing and turned into structured data that

Box 1 | **The ICD codes**

The International Classification of Diseases (ICD) codes, maintained by the WHO, are widely used in healthcare settings for record billing and clinical purposes. The ninth revision of ICD codes, ICD-9, along with its American adaption, ICD-9-CM (Clinical Modification), has been used since the late 1970s[112]. In October 2015, healthcare systems in the United States were required to transition from ICD-9 to a new coding set with greater specificity, ICD-10 (REF.[116]). To enable mapping from ICD-9 to ICD-10 in a standardized and automated manner, general equivalence mappings were developed that provide a mechanism to link ICD-9 to ICD-10, and vice versa. However, given the difference in the number of codes, many codes have no exact matches[117]. An additional challenge is that many electronic health records contain both ICD-9 and ICD-10 data due to the fairly recent mandate to switch. Developing phenotype algorithms that accommodate both sets of codes will be most informative. For further details on the differences between ICD-9 and ICD-10 codes, see REF.[118].

can be mined to help define phenotypes, such as disease status, for high-throughput phenotyping[47,48].

The patient information captured in clinical notes is likely to be greater than that in a traditional population-based study, as capturing these measures in a population-based study would be expensive and time-consuming. Nevertheless, extracting valuable information from EHRs for risk prediction remains challenging. EHR data are limited to inputs generated during clinic visits; thus, patient records contain different inputs, with sicker patients having more detailed clinical records than healthier patients. In addition, for many patients, EHR records are extremely sparse and irregularly spaced. For example, in the 670,000-individual Geisinger Health System EHR, only 20 out of 143 clinical variables had <25% missing data; more than 110 variables had >90% missing data[49]. Additionally, EHRs can only capture medications prescribed by clinicians. Whether patients adhere to taking medication as prescribed is not captured[50].

***Phenotyping algorithms.*** Complex disease phenotypes, such as T2DM[51] or cataracts[52], can be extracted using electronic phenotyping algorithms. These algorithms can include structured data, such as ICD codes, clinical procedural terminology codes, vital signs, clinical laboratory measurements and medications, in addition to unstructured data from clinical notes processed through natural language processing. For example, using a semi-supervised approach, a recent study included more than 20,000 structured and unstructured variables to infer EHR phenotypes[53].

Over 53 such algorithms that utilize combinations of clinical variables to extract phenotypes have been created and deposited in the public repository Phenotype KnowledgeBase (PheKB)[46], which maintains the pseudocode to implement these electronic algorithms across different EHRs. Phenotypes defined by these algorithms have been successfully used for GWAS. For example, the electronic Medical Records and Genomics (eMERGE) Network in the United States conducted GWAS on resistant hypertension[54] and white blood cell count[55] using an electronic phenotyping algorithm; members of the Kaiser Permanente EHR were used to identify genetic associations of intraocular pressure[56]; and cardiovascular disease and mental disorder phenotypes from the Penn Medicine BioBank have been studied for potential pleiotropic relationships[15].

These algorithms have been demonstrated to have high positive predictive value and are portable across different healthcare organizations[46]. One great strength of these phenotype algorithms is that they can be implemented in epidemiologic survey databases, such as the National Health and Nutrition Examination Survey (NHANES) and Atherosclerosis Risk in Communities (ARIC), as well as population registries, such as UK Biobank and deCODE[57]. This enables the replication or validation of a PRS in population-based cohort studies. In addition, the adoption of phenotype ontologies and/or standardized vocabularies, such as the Human Phenotype Ontology[58], Systematized Nomenclature of Medicine – Clinical Terms (SNOMED CT)[59] and LOINC (Logical Observation Identifiers Names and Codes)[60], can assist in the mapping of complex data elements collected in diverse EHRs and patient registries into more complete structured phenotype definitions.

However, these algorithms also contain significant heterogeneity. For example, there are multiple definitions of heart failure defined by different algorithms. In addition, how algorithms deal with the age-dependent nature of many phenotypes can vary[39,40,61]. Errors in classifying a phenotype can affect phenotype accuracy, which in turn can bias genetic associations[62,63] and decrease the prediction accuracy of PRSs[64]. In addition, how algorithms deal with missing data in EHRs, for example, case deletion versus imputation, is an important consideration[65].

***Machine learning.*** Recently, machine learning strategies have been applied in addition to rule-based phenotyping algorithms to improve the derivation of phenotypes that model ICD codes and other clinical variables. These machine learning approaches are emerging as a valuable alternative approach to derive EHR phenotypes[43]. For example, a recent study evaluated multiple machine learning methods (*k*-nearest neighbour, decision tree, random forest, support vector machine and naive Bayes) to mine EHR for cases of T2DM[66]. An alternative machine learning method, lasso logistic regression, was adopted to identify cases of atopic dermatitis[67]. Yet another example is the use of random forest and decision tree methods to identify cases of rheumatoid arthritis[68]. Support vector machines have also been used to define phenotype status for rheumatoid arthritis[69]. The use of machine learning is expected to continue to emerge as a powerful strategy for mining EHRs more comprehensively and gaining maximal potential from this rich phenotypic resource in the coming years[44].

## EHR-based cohort studies

In addition to the large number of phenotypes, the speed with which EHR-based research studies can be completed is an important advantage. Once a phenotype is defined, a traditional population-based cohort study requires active recruitment of study participants, phenotyping and biospecimen collection, and genotyping or sequencing of the genetic materials, which can take years from start to finish. In EHR-based studies, identification of a cohort to be studied can take a few

Pleiotropic
Pertaining to a gene that affects multiple phenotypes or traits.

Positive predictive value
The proportion of true positives among positive results.

*k*-Nearest neighbour
A machine learning method that is based on similarities between samples.

Decision tree
A machine learning method that learns decision rules from the data and represents them in a tree-like structure. The tree is used to perform classification or regression.

Random forest
An ensemble approach that learns from multiple decision trees.

Support vector machine
A supervised machine learning method that uses a hyperplane to perform classification.

Naive Bayes
A simple classification method that is based on the Bayes' theorem.

Lasso logistic regression
A penalized version of regular logistic regression. The additional penalty term forces some features to have zero coefficients.

days or a few weeks, depending on the personnel and software infrastructure that are available and their accessibility. This process can also depend on the search terms being used to identify the disease cohort[70]. Some EHRs have an established data warehouse with user-friendly search capability that makes it easy to identify a cohort, whereas other EHRs have this search capability as well as search tools developed by the EHR vendor. Cohort identification usually happens with the user blinded to personal health information data. Once a cohort is identified, the next step is to request the data. This step usually involves providing a list of fields to personnel who can access and retrieve the data securely. These data can then be cleaned and analysed. The presence of a biobank, a user-friendly data warehouse and the infrastructure for the secure handling of data can greatly reduce the time needed to go from research question to results. Thus, an EHR-based study can be completed in a matter of months, assuming the appropriate institutional review board approvals have been obtained.

Box 2 | **Challenges of PRSs**

**Sample size**

Although the mathematical formula to construct polygenic risk scores (PRSs) remains the same for electronic health record (EHR) data, some considerations are unique to EHR data. For example, the disease sample size and prevalence are not uniform in EHRs. The sample size and prevalence are influenced by the disease type, patient population and specialties of the healthcare system. Unique to EHRs, sample sizes can be different for each disease, because not all patients can be assigned to a disease status due to a lack of clinical data or uncertain diagnoses. In reality, some EHRs may have a large number of cardiovascular patients but an inadequate number of patients for other diseases, and vice versa. By contrast, there can be an adequate number of total patients but a low prevalence of cases. In these cases, the power to detect true genetic associations is diminished, which can affect the calculation of PRSs[119].

**Genotyping array versus sequencing data**

Genetic data in EHR-linked biobanks or genome-wide association studies (GWAS) can take the form of genotyping or sequencing data. Genotyping arrays generally capture common genetic variants (minor allele frequency >1%), whereas sequencing can capture both common and rare variants. If the genetic associations, or 'weights', used to calculate the PRS are generated from sequencing data, and the EHR target population's genetic data are derived from a genotyping array, only the subset of common SNPs can be used to construct the PRS for the target population. It is possible for genotyping array data to be imputed to whole-genome sequencing data to include low-frequency and rare variants[120], but the imputation effectiveness remains low for rare variants. Thus, the quality of imputation will affect the quality of the PRS.

**Transferability across populations**

In GWAS, the sample population is filtered to be more homogeneous and unrelated (generally, the identity by descent kinship coefficient is <0.05). Recently, a study showed that PRSs generated from one population had limited transferability to other populations[73]. Specifically, a height PRS generated using European cohorts predicted that African individuals were shorter than European individuals, contradicting empirical evidence[73]. Because most current GWAS participants have European ancestry, the clinical use of PRSs generated from GWAS may exacerbate health disparities[121]. A diverse EHR population may produce better PRSs by integrating multiple ethnicities[122]. To address the limited generalizability of PRSs across populations, parallel efforts are needed in both developing improved statistical methods that better utilize existing European ancestry samples and increasing the representation and sample size of populations of non-European ancestry in GWAS. There are also opportunities for method development to combine summary statistics from an EHR with GWAS data when calculating a PRS to take advantage of the specificity of the local EHR and the large number of published GWAS results. Ultimately, EHR-generated PRSs can directly benefit patients and the patient community that contributed genetic data to the EHR, establishing an additional avenue for the return of research results.

Although the creation of a biobank requires a similar level of effort to that of cohort studies as well as comparable timelines for generating the genetic data to link to the EHR, it is the longitudinal nature of data collection in the EHR where time and cost efficiencies emerge. Once genetic data have been generated, an EHR-based study can be more efficient than traditional cohort studies because researchers do not pay for the follow-up of research participants. Instead, data are routinely collected for thousands of patients' phenotypes at no extra cost to the researcher or funding agencies, and the data are widely accessible to researchers within the institution. This provides an economy of scale that cannot be matched in a typical research study. An example realization of this approach is the eMERGE Network, which is using genotype data from patient participants and carrying out association studies using phenotypes derived from EHRs[71]. Moreover, incorporating patient trajectories into research studies makes it possible to ask temporal questions. For example, a recent study from BioVU at Vanderbilt University Medical Center, which comprised over 100,000 participants, found that including longitudinal physical and laboratory measures with genetic data improved the prediction of cardiovascular disease events over the prediction models used in clinical practice[72]. Such studies can be completed using an EHR without the need for new data collection.

## Validation of an EHR-based PRS

As PRSs can be biased in multiple ways (BOX 2), proper validation to evaluate precision is important. The current practice for evaluating the generalizability of GWAS PRSs is to perform validation in independent GWAS data sets[29,73,74]. However, these cohorts may not match the patient characteristics of the initial cohort[29]. Thus, when PRSs do not show strong concordance in distinct data sets, it is difficult to pinpoint why. A key advantage of EHR-derived patient populations is that they are more representative of local healthcare system populations. Thus, a PRS generated using EHR-linked genetic data can generalize better to its patient population, assuming the population is stable. Nevertheless, EHR-generated PRSs need to also be evaluated for validity and generalizability. EHR-generated PRSs can be continuously evaluated on new patients. When PRS prediction does not agree with a patient's diagnosis, the patient's clinical record can provide additional information, such as medical history, diagnoses and medication history[75]. This enables clinicians and domain experts to manually review records to further validate predictions from PRSs. Here, the validity of PRSs is not only affected by the true underlying genetic associations but also by factors that affect the reproducibility of research findings in EHR data[76]. Thus, replication of PRSs within and across EHRs is critical for understanding the reproducibility and robustness of the construction of PRSs.

***Replication within EHR systems.*** Replication and reproducibility of polygenic risk prediction can be addressed using EHR data, as new patients are added each year, which makes it easier and less expensive to identify replication cohorts. However, applying an EHR-generated

Population stratification
The presence of allele frequency differences between subpopulations within a larger population.

Relative risk
The ratio of the probability of an event (such as disease) occurring in an at-risk group to the probability of it occurring in a population that is not considered at risk.

PRS to the record's own patients may cause overfitting of the PRS if not properly controlled. In addition, EHR patient populations may exhibit large degrees of population stratification due to the ancestry diversity of the patients who use the healthcare system. One study identified 565,000 families with a mix of ancestries from 1,890,000 patients in the Columbia University, Mount Sinai and Weill Cornell EHRs[37]. Using independent GWAS data, it has been shown that failure to control for population stratification can lead to biased estimation of PRSs[75,77].

To avoid overfitting, EHR patients who were used to generate a PRS (training samples) need to be independent of patients who use the PRS (testing samples). Several measures can be implemented to achieve this independence. PRSs can be generated using freezes of EHR data at fixed time points. Once the PRSs are generated, they will not be updated until the next data freeze. This ensures that future patients do not contribute to the generation of PRSs. Alternatively, cross-validation, where different partitions of the data are used for training and testing, can be used to separate training and testing samples for a PRS[32,78]. Under this approach, a percentage of the total patients are used to generate the PRS, and the resulting PRS is applied to the rest of the patients. Although the training and testing approach does not eliminate biases of PRS (BOX 2), any potential biases due to, for example, population structure, phenotype imprecision and confounding covariates will be consistent between patients who were used to generate PRS and patients who utilize the PRS for risk assessment. Further studies are needed to compare PRSs generated from external GWAS summary statistics versus those generated from part of an EHR's own patient population to understand and control for these EHR-specific PRS biases.

***Replication across other EHR systems.*** EHR data can be mapped to common data models such as the Observational Medical Outcomes Partnership (OMOP) standardized vocabulary[79]. Mapping EHR data to the OMOP enables the comparison of analytical results across healthcare systems, because there is confidence that the data are being mapped to the same clinical concepts. For example, the Observational Health Data Sciences and Informatics (OHDSI) initiative[80] includes dozens of sites from around the world that have agreed to adopt the OMOP standard for EHR data. This enables aggregation of research results to improve both power, using statistical methods such as the one-shot distributed algorithm (ODAL) for logistic regression[81,82], and reproducibility by assessing the replication of results across different EHRs.

As more EHRs with linked genetic data become available, PRSs generated from EHRs can be evaluated in future patients or in different EHRs, potentially with different demographic compositions. To accomplish this, both EHR infrastructures that can accommodate genetic data or genetic risk factors[83] and quantitative metrics that can properly evaluate PRS replication need to be developed, preferably in parallel. Replication of PRSs has already been shown to be influenced by uncontrolled population structure[72,77] (BOX 2). In EHRs, there are potentially more complex and often unobserved confounding factors that need to be evaluated in the context of PRSs[84].

## Integrating data

Data integration is becoming increasingly common for genetic studies with more collaborative consortia, such as the eMERGE Network and the NIH All of Us initiative[85]. Proper integration of data across sites requires consistent phenotype definitions and quality control procedures. One approach to ensure consistency is through the use of common data models that harmonize input from different EHRs. The Exploring and Understanding Adverse Drug Reactions by integrative mining of clinical records and biomedical knowledge (EU-ADR) project utilized common data models to combine data from multiple European EHRs to monitor drug safety[86,87]. Similarly, common data models have been applied to EHRs from Asia[88], Canada, the United States and Europe for pharmacoepidemiology studies[89].

Integrating data across sites can potentially accelerate knowledge discovery and enhance the generalizability of research findings. For PRSs of rare disease, integrating data from different EHRs enables larger pooled sample sizes and needed statistical power to identify genetic risk factors that cannot be revealed by a single study. In some situations, directly pooling patient-level genetic data and EHR data across sites is not feasible due to logistic reasons and patient privacy concerns. In this context, there are multiple strategies, in a meta-analytic form, that can construct PRSs based on summary statistics (FIG. 1). However, for rare diseases, integration methods based on summary statistics are limited, as pooled estimates can be biased[90]. Alternatively, distributed computing algorithms, which decompose a unified computational task into tasks that can be carried out within each site without sharing patient-level data[81,91–94], can be considered.

## Risk estimation beyond a PRS

Genetics is merely one contributing factor to disease susceptibility. Furthermore, a PRS only captures a portion of total genetic heritability[95]. Thus, PRSs alone can be used only to evaluate the relative risk of diseases[24]. The area under the curve (AUC) and the coefficient of determination ($R^2$) are the most common metrics for evaluating PRS prediction of binary and continuous phenotypes, respectively[31]. However, there are several other accuracy metrics, such as positive and negative predictive values, the F1 score, Hosmer–Lemeshow goodness of fit, the Brier score, the odds ratio and root mean squared errors, that can be used to evaluate the performance of a PRS[12,32,96]. No one metric can capture the entire discrimination, calibration, prediction accuracy and risk stratification properties of a PRS. For example, a study has shown that PRSs are the most informative for patients in the top percentile of the PRS distribution[35]; the top 8, 6.1, 3.5, 3.2 and 1.5 percentiles of a PRS had a more than threefold increased risk of coronary artery disease, atrial fibrillation, T2DM, inflammatory bowel disease and breast cancer, respectively[35]. The commonly used area under the curve would not capture the enrichment of high-risk patients in the top percentile of the PRS.

Absolute risk
The actual probability of disease occurrence.

Omnigenic model
A model that proposes that the genetic architecture of a trait or disease is affected by many genes.

To evaluate absolute risk, much more patient information is needed, including age-dependent disease prevalence, environmental exposures and family histories. Thus, PRSs should be modelled as just one of the factors that contribute to disease risk. For example, the iCARE package jointly models multiple data sources that provide relative risks, risk factor distributions and age-dependent disease prevalence through the Cox proportional hazard model to obtain absolute risk[97,98]. Genetic data, in the form of individual SNPs or a PRS, can be modelled as one of the risk factors in iCARE, similar to other risk factors.

As FIG. 3 shows, GWAS data can have multiple data elements that are necessary pieces of information to evaluate absolute disease risk. GWAS data generally contain patient demographic variables and a selected number of environmental and lifestyle variables. Both GWAS and genetic-linked EHR data include genetic risk factors in the form of individual SNPs or PRSs, but an EHR can contain additional information on the patients' clinical history, family history and medications, more detailed environmental exposures and self-reported lifestyle variables. The additional information in EHRs allows the estimated disease risk to better approximate the absolute risk. One of the key components in estimating the absolute disease risk is to have an accurate estimate of baseline risk, which is proportional to disease prevalence. EHR data can provide a better estimate of the baseline risk compared with national survey or epidemiologic data, because the EHR patient composition can deviate from national averages due to local demographics, healthcare specialties and other factors that correlate with recorded clinic visits[99]. The baseline risk can be incorporated into the PRS prediction model to better predict the disease risk of a 'biased' patient population[97].

## Conclusions and future directions

The utility of PRSs developed using GWAS data has been extensively demonstrated for many common diseases. By contrast, repurposing EHRs with linked genetic data for PRS prediction is still at an early stage. In this Review, we have discussed several advantages of using EHR data for developing a PRS, including easily accessible rich sets of phenotypes, the potential for both internal and external validation, and the diversity of the EHR patient population. Nevertheless, dealing with EHR data poses many challenges that should be thoroughly considered when implementing EHR-based PRSs, including phenotype extraction from clinical records, the complex population structures and relatedness that are common in EHR patient populations, and the potential to overfit a PRS for a local population.

In the United States, the adoption of EHRs has reached over 86% as of 2017, according to the Office of the National Coordinator for Health Information Technology. However, EHRs that have linked patient genetic data are still in the minority. More EHRs with larger samples of linked genetic data are needed to take full advantage of the clinical data in EHRs. When using EHR data, patients' privacy should be paramount. However, challenges arise when using and sharing patients' genetic and clinical data within or across multiple EHRs. In particular, disease risk prediction based on patient characteristics, including PRSs, has the potential to be used against the patient, and thus tight regulation is needed. To balance the advantage of advancing healthcare using large-scale EHR data and potential concerns of privacy violations, more up-to-date regulatory measures are needed to match the pace of technological development[100].

Genetics-linked EHR data can offer a unique opportunity to systematically evaluate the omnigenic model across hundreds or thousands of phenotypes[101]. In one study, increasing $P$-value thresholds ranging from $5\times10^{-8}$ to 1 were used to select SNPs and their corresponding PRS performances were compared. The most predictive PRS included from 5,158 SNPs for breast cancer to 6,893,037 SNPs for T2DM[35]. The performance of a PRS could potentially reflect the degree of omnigenicity of a phenotype[102].

Armed with the environmental variables recorded in EHRs, PRSs can be used for gene and environment interaction studies, that is, a PRS and the environment[103]. For example, it would be interesting to evaluate the predictability of PRSs under different environmental conditions, such as PRS in smoking versus non-smoking populations, although the interpretation of these studies would be challenging given the large number of SNPs included in PRSs. Of note, although EHRs contain more risk factor variables than GWAS, they are still limited in scope. Although some environmental data can be derived from patients' addresses, enabling the inclusion of census information, such as socio-economic status, and environmental measures, such as air pollution, other potentially informative day-to-day environmental exposures and lifestyle factors are yet to be included in EHRs.

More comprehensive validation of both the PRS methodology and results is needed before this approach can be implemented in the clinical setting for evaluating disease risk. Besides the current practice of validating PRSs in independent data sets, it is crucial to enlist clinicians to evaluate the PRS in healthcare settings, where considerably more information about patients is stored. Although clinician engagement also occurs in cohort studies, there are several important reasons to engage clinicians for the utilization of PRSs in EHR systems. First, clinicians will have first-hand knowledge of the patient's clinical history and can have important insights into the definition of the phenotypes of interest. Second, they have insight into which data should be studied as well as their limitations and quality control issues. Third, they can help to interpret and communicate results. Fourth, they can often suggest questions that are clinically relevant. Last, collaborating physicians can facilitate the implementation of a PRS-based risk model in the clinic for validation and routine use.

It will also be important to determine the conditions under which PRSs do not give clinically relevant predictions or contradict a clinician's diagnosis and understand the reasons why. For example, ethnic disparities in GWAS data have limited the generalizability of PRSs in the overall population[104]. Given the limited resources of EHR systems, they should be incentivized to include

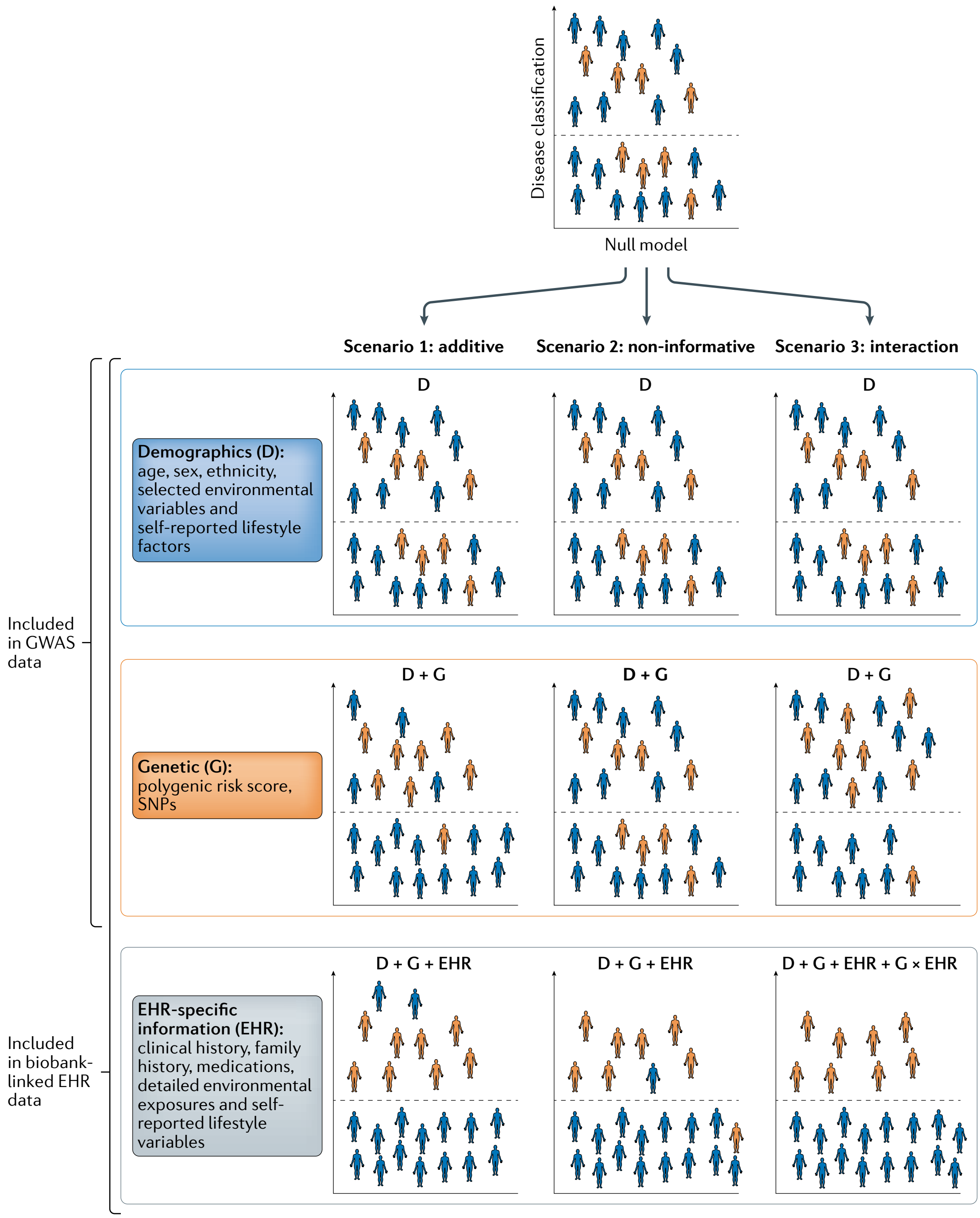


Fig. 3 | **Risk prediction in EHR data using the PRS with other clinical factors.** Under the null model, patients with disease (orange) and controls (blue) are randomly distributed across the classification threshold. Information on patients' demographics, genetics and variables stored in electronic health records (EHRs) can potentially improve risk prediction in multiple ways. For example, in scenario 1, the improvement of prediction is additive in that each type of data independently improves disease classification. In scenario 2, demographics and variables in EHRs contribute to the improvement of disease classification. However, some information, such as the polygenic risk score (PRS), can be non-informative because it does not improve risk prediction. In scenario 3, the marginal effect of genetic factors can slightly improve classification accuracy, but the interaction between genetic factors and EHR variables can significantly improve it. GWAS, genome-wide association study; SNP, single-nucleotide polymorphism.

more ethnic minorities for genetic sequencing so that improvements in PRSs are not only quantitative in terms of sample sizes but also qualitative, to improve risk prediction for the entire population. This will require concerted efforts between geneticists, methodologists and domain experts to successfully utilize PRSs for disease prevention and treatment.

In a broad sense, the validity of a PRS, the added clinical value of a PRS and the ethical, legal and social issues of implementing a PRS in different populations all remain to be rigorously evaluated[105]. For a particular phenotype, the allocation of therapies based on PRSs can be influenced by disease prevalence, the relative risk of the top-percentile PRS group and the precision and efficacy of available therapeutics[106]. Thus, the exact clinical utility of a PRS will vary from disease to disease. Notably, for cancers that have strong genetic susceptibility, the PRS has already been explored as a risk factor in European populations[107,108]. For many traits for which PRSs are being generated, clinical guidelines have already been published[109,110]. Efforts are underway to evaluate how to integrate the PRS with these known clinical risk factors to build genomic risk assessments. Future work is needed to integrate genetic risk factors, such as PRSs, risk factors from clinical data within EHRs and risk factors from external sources, to achieve a comprehensive assessment of disease risk.

Published online: 31 March 2020


1. Preiss, D. & Kristensen, S. L. The new pooled cohort equations risk calculator. *Can. J. Cardiol.* **31**, 613–619 (2015).
2. Antoniou, A. et al. Average risks of breast and ovarian cancer associated with BRCA1 or BRCA2 mutations detected in case series unselected for family history: a combined analysis of 22 studies. *Am. J. Hum. Genet.* **72**, 1117–1130 (2003).
3. O'Sullivan, B. P. & Freedman, S. D. Cystic fibrosis. *Lancet* **373**, 1891–1904 (2009).
4. Manolio, T. A. et al. Finding the missing heritability of complex diseases. *Nature* **461**, 747–753 (2009).
5. Lo, A., Chernoff, H., Zheng, T. & Lo, S.-H. Why significant variables aren't automatically good predictors. *Proc. Natl Acad. Sci. USA* **112**, 13892–13897 (2015).
6. Visscher, P. M. et al. 10 Years of GWAS discovery: biology, function, and translation. *Am. J. Hum. Genet.* **101**, 5–22 (2017).
7. Bogdan, R., Baranger, D. A. A. & Agrawal, A. Polygenic risk scores in clinical psychology: bridging genomic risk to individual differences. *Annu. Rev. Clin. Psychol.* **14**, 119–157 (2018).
8. Schizophrenia Working Group of the Psychiatric Genomics Consortium. Biological insights from 108 schizophrenia-associated genetic loci. *Nature* **511**, 421–427 (2014).
9. Zhang, J.-P. et al. Schizophrenia polygenic risk score as a predictor of antipsychotic efficacy in first-episode psychosis. *Am. J. Psychiatry* **176**, 21–28 (2019).
10. Jones, H. J. et al. Phenotypic manifestation of genetic risk for schizophrenia during adolescence in the general population. *JAMA Psychiatry* **73**, 221 (2016).
11. Kohane, I. S. Using electronic health records to drive discovery in disease genomics. *Nat. Rev. Genet.* **12**, 417–428 (2011).
12. Fritsche, L. G. et al. Association of polygenic risk scores for multiple cancers in a phenome-wide study: results from the michigan genomics initiative. *Am. J. Hum. Genet.* **102**, 1048–1061 (2018).
**This analysis uses biobank-linked EHR data to study PRS associations with cancers.**
13. Banda, Y. et al. Characterizing race/ethnicity and genetic ancestry for 100,000 subjects in the Genetic Epidemiology Research on Adult Health and Aging (GERA) cohort. *Genetics* **200**, 1285–1295 (2015).
14. Kvale, M. N. et al. Genotyping informatics and quality control for 100,000 subjects in the Genetic Epidemiology Research on Adult Health and Aging (GERA) cohort. *Genetics* **200**, 1051–1060 (2015).
15. Li, R. et al. A regression framework to uncover pleiotropy in large-scale electronic health record data. *J. Am. Med. Informatics Assoc* **26**, 1083–1090 (2019).
16. McCarty, C. A., Wilke, R. A., Giampietro, P. F., Wesbrook, S. D. & Caldwell, M. D. Marshfield Clinic Personalized Medicine Research Project (PMRP): design, methods and recruitment for a large population-based biobank. *Per. Med.* **2**, 49–79 (2005).
17. Nagai, A. et al. Overview of the Biobank Japan project: study design and profile. *J. Epidemiol.* **27**, S2–S8 (2017).
18. Cho, S. Y. et al. Opening of the National Biobank of Korea as the infrastructure of future biomedical science in Korea. *Osong Public. Heal. Res. Perspect.* **3**, 177–184 (2012).
19. Chen, Z. et al. China Kadoorie Biobank of 0.5 million people: survey methods, baseline characteristics and long-term follow-up. *Int. J. Epidemiol.* **40**, 1652–1666 (2011).
20. Locke, A. E. et al. Exome sequencing of Finnish isolates enhances rare-variant association power. *Nature* **572**, 323–328 (2019).
21. Sankar, P. L. & Parker, L. S. The Precision Medicine Initiative's all of us research program: an agenda for research on its ethical, legal, and social issues. *Genet. Med.* **19**, 743–750 (2017).
22. Bycroft, C. et al. The UK Biobank resource with deep phenotyping and genomic data. *Nature* **562**, 203–209 (2018).
**This paper presents one of the largest genetic-linked patient clinical data sets that is publicly available to researchers.**
23. Casey, J. A., Schwartz, B. S., Stewart, W. F. & Adler, N. E. Using electronic health records for population health research: a review of methods and applications. *Annu. Rev. Public Health* **37**, 61–81 (2016).
24. Chatterjee, N., Shi, J. & García-Closas, M. Developing and evaluating polygenic risk prediction models for stratified disease prevention. *Nat. Rev. Genet.* **17**, 392–406 (2016).
**This review article provides an overview of risk prediction methods and approaches to incorporate a PRS into risk models.**
25. Torkamani, A., Wineinger, N. E. & Topol, E. J. The personal and clinical utility of polygenic risk scores. *Nat. Rev. Genet.* **19**, 581–590 (2018).
**This paper presents a review of the background of the PRS and how it can be utilized for risk predictions.**
26. Li, R., Chen, Y. & Moore, J. H. Integration of genetic and clinical information to improve imputation of data missing from electronic health records. *J. Am. Med. Informatics Assoc.* **26**, 1056–1063 (2019).
27. Shi, J. et al. Winner's curse correction and variable thresholding improve performance of polygenic risk modeling based on genome-wide association study summary-level data. *PLoS Genet.* **12**, e1006493 (2016).
28. Dudbridge, F. Power and predictive accuracy of polygenic risk scores. *PLoS Genet.* **9**, e1003348 (2013).
29. Vilhjálmsson, B. J. et al. Modeling linkage disequilibrium increases accuracy of polygenic risk scores. *Am. J. Hum. Genet.* **97**, 576–592 (2015).
**This study shows that the accuracy of a PRS is affected by the modelling of linkage disequilibrium between SNPs.**
30. Mak, T. S. H., Porsch, R. M., Choi, S. W., Zhou, X. & Sham, P. C. Polygenic scores via penalized regression on summary statistics. *Genet. Epidemiol.* **41**, 469–480 (2017).
31. Choi, S. W., Mak, T. S. H. & O'Reilly, P. F. A guide to performing polygenic risk score analyses. Preprint at *bioRxiv* https://doi.org/10.1101/416545 (2018).
32. Fritsche, L. G. et al. Exploring various polygenic risk scores for skin cancer in the phenomes of the Michigan Genomics Initiative and the UK Biobank with a visual catalog: PRSWeb. *PLoS Genet.* **15**, e1008202 (2019).
33. Reus, L. M. et al. Association of polygenic risk for major psychiatric illness with subcortical volumes and white matter integrity in UK Biobank. *Sci. Rep.* **7**, 42140 (2017).
34. Mavaddat, N. et al. Polygenic risk scores for prediction of breast cancer and breast cancer subtypes. *Am. J. Hum. Genet.* **104**, 21–34 (2019).
35. Khera, A. V. et al. Genome-wide polygenic scores for common diseases identify individuals with risk equivalent to monogenic mutations. *Nat. Genet.* **50**, 1219–1224 (2018).
**This study demonstrates that a PRS can identify individuals who have a clinically significantly increased risk of coronary artery disease, atrial fibrillation, T2DM, inflammatory bowel disease and breast cancer.**
36. Khera, A. V. et al. Polygenic prediction of weight and obesity trajectories from birth to adulthood. *Cell* **177**, 587–596.e9 (2019).
37. Polubriaginof, F. C. G. et al. Disease heritability inferred from familial relationships reported in medical records. *Cell* **173**, 1692–1704.e11 (2018).
38. DeBoever, C. et al. Assessing digital phenotyping to enhance genetic studies of human diseases. Preprint at *bioRxiv* https://doi.org/10.1101/738856 (2019).
39. Robinson, J. R., Wei, W.-Q., Roden, D. M. & Denny, J. C. Defining phenotypes from clinical data to drive genomic research. *Annu. Rev. Biomed. Data Sci.* **1**, 69–92 (2018).
40. Wei, W.-Q. & Denny, J. C. Extracting research-quality phenotypes from electronic health records to support precision medicine. *Genome Med.* **7**, 41 (2015).
41. Denny, J. C. et al. PheWAS: demonstrating the feasibility of a phenome-wide scan to discover gene–disease associations. *Bioinformatics* **26**, 1205–1210 (2010).
42. Chiu, P.-H. & Hripcsak, G. EHR-based phenotyping: bulk learning and evaluation. *J. Biomed. Inform.* **70**, 35–51 (2017).
43. Banda, J. M., Seneviratne, M., Hernandez-Boussard, T. & Shah, N. H. Advances in electronic phenotyping: from rule-based definitions to machine learning models. *Annu. Rev. Biomed. Data Sci.* **1**, 53–68 (2018).
44. Ritchie, M. D. Large-scale analysis of genetic and clinical patient data. *Annu. Rev. Biomed. Data Sci.* **1**, 263–274 (2018).
45. Hripcsak, G. & Albers, D. J. Next-generation phenotyping of electronic health records. *J. Am. Med. Inform. Assoc.* **20**, 117–121 (2013).
46. Kirby, J. C. et al. PheKB: a catalog and workflow for creating electronic phenotype algorithms for transportability. *J. Am. Med. Inform. Assoc.* **23**, 1046–1052 (2016).
**This paper discusses PheKB, which contains a wide range of phenotyping algorithms that can automatically extract phenotypes from EHR data.**
47. Liao, K. P. et al. Development of phenotype algorithms using electronic medical records and incorporating natural language processing. *BMJ* **350**, h1885 (2015).
48. Yu, S. et al. Surrogate-assisted feature extraction for high-throughput phenotyping. *J. Am. Med. Inform. Assoc.* **24**, e143–e149 (2017).
49. Beaulieu-Jones, B. K. et al. Characterizing and managing missing structured data in electronic health records: data analysis. *JMIR Med. Inform.* **6**, e11 (2018).
50. Kleinsinger, F. The unmet challenge of medication nonadherence. *Perm. J.* **22**, 18-033 (2018).
51. Kho, A. N. et al. Use of diverse electronic medical record systems to identify genetic risk for type 2 diabetes within a genome-wide association study. *J. Am. Med. Inform. Assoc.* **19**, 212–218 (2012).
52. Peissig, P. L. et al. Importance of multi-modal approaches to effectively identify cataract cases from electronic health records. *J. Am. Med. Inform. Assoc.* **19**, 225–234 (2012).
53. Halpern, Y., Horng, S., Choi, Y. & Sontag, D. Electronic medical record phenotyping using the anchor and learn framework. *J. Am. Med. Inform. Assoc.* **23**, 731–740 (2016).
54. Dumitrescu, L. et al. Genome-wide study of resistant hypertension identified from electronic health records. *PLoS One* **12**, e0171745 (2017).
55. Crosslin, D. R. et al. Genetic variants associated with the white blood cell count in 13,923 subjects in the eMERGE Network. *Hum. Genet.* **131**, 639–652 (2012).
56. Choquet, H. et al. A large multi-ethnic genome-wide association study identifies novel genetic loci for intraocular pressure. *Nat. Commun.* **8**, 2108 (2017).

57. Gudbjartsson, D. F. et al. Large-scale whole-genome sequencing of the Icelandic population. *Nat. Genet.* **47**, 435–444 (2015).
58. Robinson, P. N. et al. The Human Phenotype Ontology: a tool for annotating and analyzing human hereditary disease. *Am. J. Hum. Genet.* **83**, 610–615 (2008).
59. Randorff Højen, A. & Rosenbeck Gøeg, K. SNOMED CT implementation. *Methods Inf. Med.* **51**, 529–538 (2012).
60. Vreeman, D. J., McDonald, C. J. & Huff, S. M. LOINC®: a universal catalogue of individual clinical observations and uniform representation of enumerated collections. *Int. J. Funct. Inform. Personal. Med.* **3**, 273 (2010).
61. Schulam, P., Wigley, F. & Saria, S. Clustering longitudinal clinical marker trajectories from electronic health data: applications to phenotyping and endotype discovery. *Proc. Natl Conf. Artif. Intell.* **4**, 2956–2964 (2015).
62. Duan, R. et al. An empirical study for impacts of measurement errors on EHR based association studies. *AMIA Annu. Symp. Proc.* **2016**, 1764–1773 (2017).
63. Chen, Y., Wang, J., Chubak, J. & Hubbard, R. A. Inflation of type I error rates due to differential misclassification in EHR-derived outcomes: empirical illustration using breast cancer recurrence. *Pharmacoepidemiol. Drug Saf.* **28**, 264–268 (2019).
64. Li, R., Tong, J., Duan, R., Chen, Y. & Moore, J. H. Evaluation of phenotyping errors on polygenic risk score predictions. *Proc. Int. Joint Conf. Biomed. Eng. Syst. Technol.* https://doi.org/10.5220/0008935301230130 (2020).
65. Wells, B. J., Chagin, K. M., Nowacki, A. S. & Kattan, M. W. Strategies for handling missing data in electronic health record derived data. *EGEMS* **1**, 1035 (2013).
66. Zheng, T. et al. A machine learning-based framework to identify type 2 diabetes through electronic health records. *Int. J. Med. Inform.* **97**, 120–127 (2017).
67. Gustafson, E., Pacheco, J., Wehbe, F., Silverberg, J. & Thompson, W. A machine learning algorithm for identifying atopic dermatitis in adults from electronic health records. *IEEE Int. Conf. Healthc. Inform.* **2017**, 83–90 (2017).
68. Zhou, S.-M. et al. Defining disease phenotypes in primary care electronic health records by a machine learning approach: a case study in identifying rheumatoid arthritis. *PLoS One* **11**, e0154515 (2016).
69. Carroll, R. J., Eyler, A. E. & Denny, J. C. Naïve electronic health record phenotype identification for rheumatoid arthritis. *AMIA Annu. Symp. Proc.* **2011**, 189–196 (2011).
70. Cimino, J. J., Lancaster, W. J. & Wyatt, M. C. Classification of clinical research study eligibility criteria to support multi-stage cohort identification using clinical data repositories. *Stud. Health Technol. Inform.* **245**, 341–345 (2017).
71. Gottesman, O. et al. The electronic Medical Records and Genomics (eMERGE) Network: past, present, and future. *Genet. Med.* **15**, 761–771 (2013).
72. Zhao, J. et al. Learning from longitudinal data in electronic health record and genetic data to improve cardiovascular event prediction. *Sci. Rep.* **9**, 717 (2019).
73. Martin, A. R. et al. Human demographic history impacts genetic risk prediction across diverse populations. *Am. J. Hum. Genet.* **100**, 635–649 (2017).
74. Sohail, M. et al. Polygenic adaptation on height is overestimated due to uncorrected stratification in genome-wide association studies. *eLife* **8**, 39702 (2019).
75. Zeng, Z., Deng, Y., Li, X., Naumann, T. & Luo, Y. Natural language processing for EHR-based computational phenotyping. *IEEE/ACM Trans. Comput. Biol. Bioinform.* **16**, 139–153 (2019).
76. Denaxas, S. et al. Methods for enhancing the reproducibility of biomedical research findings using electronic health records. *BioData Min.* **10**, 31 (2017).
77. Berg, J. J. et al. Reduced signal for polygenic adaptation of height in UK Biobank. *eLife* **8**, e39725 (2019).
78. Gao, X. R., Huang, H. & Kim, H. Polygenic risk score is associated with intraocular pressure and improves glaucoma prediction in the UK Biobank cohort. *Transl. Vis. Sci. Technol.* **8**, 10 (2019).
79. Stang, P. E. et al. Advancing the science for active surveillance: rationale and design for the observational medical outcomes partnership. *Ann. Intern. Med.* **153**, 600 (2010).
80. Hripcsak, G. et al. Observational Health Data Sciences and Informatics (OHDSI): opportunities for observational researchers. *Stud. Health Technol. Inform.* **216**, 574–578 (2015).
81. Duan, R., Boland, M. R., Moore, J. H. & Chen, Y. ODAL: a one-shot distributed algorithm to perform logistic regressions on electronic health records data from multiple clinical sites. *Pac. Symp. Biocomput.* **24**, 30–41 (2019).
82. Duan, R. et al. Learning from electronic health records across multiple sites: a communication-efficient and privacy-preserving distributed algorithm. *J. Am. Med. Informatics Assoc.* **27**, 376–385 (2019).
83. Ohno-Machado, L., Kim, J., Gabriel, R. A., Kuo, G. M. & Hogarth, M. A. Genomics and electronic health record systems. *Hum. Mol. Genet.* **27**, R48–R55 (2018).
84. Farmer, R. et al. Promises and pitfalls of electronic health record analysis. *Diabetologia* **61**, 1241–1248 (2018).
85. Denny, J. C. et al. The "All of Us" research program. *N. Engl. J. Med.* **381**, 668–676 (2019).
86. Coloma, P. M. et al. Combining electronic healthcare databases in Europe to allow for large-scale drug safety monitoring: the EU-ADR project. *Pharmacoepidemiol. Drug Saf.* **20**, 1–11 (2011).
87. Trifiro, G. et al. The EU-ADR project: preliminary results and perspective. *Stud. Health Technol. Inform.* **148**, 43–49 (2009).
88. Lai, E. C.-C. et al. Applying a common data model to Asian databases for multinational pharmacoepidemiologic studies: opportunities and challenges. *Clin. Epidemiol.* **10**, 875–885 (2018).
89. Platt, R. W. et al. How pharmacoepidemiology networks can manage distributed analyses to improve replicability and transparency and minimize bias. *Pharmacoepidemiol. Drug Saf.* **29**, 3–7 (2019).
90. Greco, T., Zangrillo, A., Biondi-Zoccai, G. & Landoni, G. Meta-analysis: pitfalls and hints. *Heart Lung Vessel.* **5**, 219–225 (2013).
91. Lu, C.-L. et al. WebDISCO: a web service for distributed Cox model learning without patient-level data sharing. *J. Am. Med. Inform. Assoc.* **22**, ocv083 (2015).
92. Wu, Y., Jiang, X., Kim, J. & Ohno-Machado, L. Grid Binary LOgistic REgression (GLORE): building shared models without sharing data. *J. Am. Med. Inform. Assoc.* **19**, 758–764 (2012).
93. Yixin Chen et al. Regression cubes with lossless compression and aggregation. *IEEE Trans. Knowl. Data Eng.* **18**, 1585–1599 (2006).
94. Wang, J., Kolar, M., Srebro, N. & Zhang, T. Efficient distributed learning with sparsity. Preprint at *arXiv* https://arxiv.org/abs/1605.07991 (2016).
95. Wray, N. R., Kemper, K. E., Hayes, B. J., Goddard, M. E. & Visscher, P. M. Complex trait prediction from genome data: contrasting EBV in livestock to PRS in humans. *Genetics* **211**, 1131–1141 (2019).
96. Powers, D. M. W. Evaluation: from precision, recall and F-factor to ROC, informedness, markedness & correlation. *J. Mach. Learn. Technol.* **2**, 37–63 (2011).
97. Choudhury, P. P. et al. iCARE: an R package to build, validate and apply absolute risk models. *PLoS One* **15**, e0228198 (2020).
98. Choudhury, P. P. et al. Comparative validation of breast cancer risk prediction models and projections for future risk stratification. *J. Natl. Cancer Inst.* **112**, 278–285 (2019).
99. Violán, C. et al. Comparison of the information provided by electronic health records data and a population health survey to estimate prevalence of selected health conditions and multimorbidity. *BMC Public Health* **13**, 251 (2013).
100. Price, W. N. & Cohen, I. G. Privacy in the age of medical big data. *Nat. Med.* **25**, 37–43 (2019).
101. Boyle, E. A., Li, Y. I. & Pritchard, J. K. An expanded view of complex traits: from polygenic to omnigenic. *Cell* **169**, 1177–1186 (2017).
102. Rammos, A., Gonzalez, L. A. N., Weinberger, D. R., Mitchell, K. J. & Nicodemus, K. K. The role of polygenic risk score gene-set analysis in the context of the omnigenic model of schizophrenia. *Neuropsychopharmacology* **44**, 1562–1569 (2019).
103. Meisner, A., Kundu, P. & Chatterjee, N. Case-only analysis of gene–environment interactions using polygenic risk scores. *Am. J. Epidemiol.* **188**, 2013–2020 (2019).
104. Manolio, T. A. Using the data we have: improving diversity in genomic research. *Am. J. Hum. Genet.* **105**, 233–236 (2019).
105. Khoury, M. J. & Mensah, G. A. Is it time to integrate polygenic risk scores into clinical practice? Let's do the science first and follow the evidence wherever it takes us! *CDC* https://blogs.cdc.gov/genomics/2019/06/03/is-it-time/ (2019)
106. Gibson, G. On the utilization of polygenic risk scores for therapeutic targeting. *PLoS Genet.* **15**, e1008060 (2019).
107. Lee, A. et al. BOADICEA: a comprehensive breast cancer risk prediction model incorporating genetic and nongenetic risk factors. *Genet. Med.* **21**, 1708–1718 (2019).
**This study integrates known cancer genes, a PRS, lifestyle risk factors and mammographic density to better estimate breast cancer risk in women.**
108. Pashayan, N. et al. Reducing overdiagnosis by polygenic risk-stratified screening: findings from the Finnish section of the ERSPC. *Br. J. Cancer* **113**, 1086–1093 (2015).
109. Lambert, S. A., Abraham, G. & Inouye, M. Towards clinical utility of polygenic risk scores. *Hum. Mol. Genet.* **28**, R133–R142 (2019).
110. Arnett, D. K. et al. 2019 ACC/AHA guideline on the primary prevention of cardiovascular disease: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J. Am. Coll. Cardiol.* **74**, e177–e232 (2019).
111. Bielinski, S. J. & Pathak, J. Heart failure with differentiation between reduced and preserved ejection fraction — Phenotype algorithm pseudo code (Mayo Clinic). *NCBI* https://www.ncbi.nlm.nih.gov/projects/gap/cgi-bin/GetPdf.cgi?id=phd004988.1 (2014)
112. National Center for Health Statistics & Centers for Disease Control and Prevention. International classification of diseases, ninth revision (ICD-9) (CDC, 1998).
113. Côté, R. A. & Robboy, S. Progress in medical information management. *JAMA* **243**, 756 (1980).
114. McDonald, C. J. et al. LOINC, a universal standard for identifying laboratory observations: a 5-year update. *Clin. Chem.* **49**, 624–633 (2003).
115. Fung, K. W., McDonald, C. & Bray, B. E. RxTerms — a drug interface terminology derived from RxNorm. *AMIA Annu. Symp. Proc.* **2008**, 227–231 (2008).
116. ICD.Codes. The switch from ICD-9 to ICD-10: when and why. *ICD.Codes* https://icd.codes/articles/icd9-to-icd10-explained (2015)
117. Topaz, M., Shafran-Topaz, L. & Bowles, K. H. ICD-9 to ICD-10: evolution, revolution, and current debates in the United States. *Perspect. Heal. Inf. Manag.* **10**, 1d (2013).
118. American Medical Association. Preparing for the ICD-10 code set: the differences between ICD-9 and ICD-10 (AMA, 2014)
119. Hong, E. P. & Park, J. W. Sample size and statistical power calculation in genetic association studies. *Genomics Inform.* **10**, 117–122 (2012).
120. Mitt, M. et al. Improved imputation accuracy of rare and low-frequency variants using population-specific high-coverage WGS-based imputation reference panel. *Eur. J. Hum. Genet.* **25**, 869–876 (2017).
121. Martin, A. R. et al. Clinical use of current polygenic risk scores may exacerbate health disparities. *Nat. Genet.* **51**, 584–591 (2019).
**This study demonstrates the lack of transferability of a PRS across different populations.**
122. Márquez-Luna, C., Loh, P.-R., South Asian Type 2 Diabetes (SAT2D) Consortium, SIGMA Type 2 Diabetes Consortium & Price, A. L. Multiethnic polygenic risk scores improve risk prediction in diverse populations. *Genet. Epidemiol.* **41**, 811–823 (2017).
**This study uses data from multiple populations to derive a more generalizable PRS for T2DM.**



### Acknowledgements
This work was supported by National Institutes of Health grants LM010098 and AI116794.


### Author contributions
The authors contributed equally to all aspects of the article.

### Competing interests
The authors declare no competing interests.

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